

# Notes: Fagereng, Gottlieb, and Guiso (JF, 2017)

## Asset Market Participation and Portfolio Choice over the Life-Cycle

### Contents

|          |                                                                             |           |
|----------|-----------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Big picture</b>                                                          | <b>2</b>  |
| <b>2</b> | <b>Data and measurement (key objects)</b>                                   | <b>3</b>  |
| 2.1      | Norwegian administrative portfolio data . . . . .                           | 3         |
| 2.2      | Assets and portfolio definitions . . . . .                                  | 4         |
| 2.3      | Key descriptive facts from Table I . . . . .                                | 4         |
| 2.4      | Life-cycle objects measured in the paper . . . . .                          | 4         |
| 2.5      | Unobservables and timing . . . . .                                          | 5         |
| <b>3</b> | <b>Models and empirical specifications (all equations + interpretation)</b> | <b>5</b>  |
| 3.1      | Empirical selection model for participation and risky share . . . . .       | 5         |
| 3.2      | Age, time, and cohort identification . . . . .                              | 6         |
| 3.3      | Financial wealth profile . . . . .                                          | 6         |
| 3.4      | Household objective in the structural model . . . . .                       | 7         |
| 3.5      | Financial markets and disaster risk . . . . .                               | 7         |
| 3.6      | Budget constraints and labor income . . . . .                               | 8         |
| 3.7      | Bellman equations and participation decision . . . . .                      | 8         |
| 3.8      | Bequest motive . . . . .                                                    | 9         |
| 3.9      | Structural estimation by simulated moments . . . . .                        | 9         |
| <b>4</b> | <b>Identification and instruments (what makes the estimates credible)</b>   | <b>10</b> |
| 4.1      | What is hard to identify . . . . .                                          | 10        |
| 4.2      | What makes the data credible . . . . .                                      | 10        |
| 4.3      | What identifies the empirical life-cycle profiles . . . . .                 | 10        |
| 4.4      | What identifies the structural parameters . . . . .                         | 11        |

|          |                                                                                      |           |
|----------|--------------------------------------------------------------------------------------|-----------|
| 4.5      | Empirical credibility and limitations . . . . .                                      | 11        |
| 4.6      | Relation to the literature . . . . .                                                 | 11        |
| <b>5</b> | <b>Tables and figures</b>                                                            | <b>12</b> |
| 5.1      | Table I: Descriptive statistics . . . . .                                            | 12        |
| 5.2      | Figure 1: Participation rates by cohort . . . . .                                    | 12        |
| 5.3      | Table II and Figure 2: Entry and exit . . . . .                                      | 13        |
| 5.4      | Figure 3 and Figure 4: Risky share and financial wealth by cohort . . . . .          | 13        |
| 5.5      | Table III and Figure 5: Heckman estimates of participation and risky share . . . . . | 14        |
| 5.6      | Figure 6 and Figure 7: Entry, exit, and wealth profiles . . . . .                    | 14        |
| 5.7      | Table IV: Parameterization . . . . .                                                 | 15        |
| 5.8      | Figure 8 and Figure 9: Policy functions and participation thresholds . . . . .       | 15        |
| 5.9      | Figure 10: Model simulations . . . . .                                               | 16        |
| 5.10     | Table V: Structural estimation . . . . .                                             | 17        |
| 5.11     | Figure 11 and Figure 12: Structural fit . . . . .                                    | 17        |
| 5.12     | Appendix tables: Labor income and crash calibration . . . . .                        | 18        |
| <b>6</b> | <b>Short summary</b>                                                                 | <b>18</b> |
| <b>7</b> | <b>Conclusion</b>                                                                    | <b>19</b> |
| 7.1      | Core conclusion . . . . .                                                            | 19        |
| 7.2      | Economic intuition . . . . .                                                         | 19        |
| 7.3      | Takeaway . . . . .                                                                   | 19        |

## 1 Big picture

- **Question.** How do households adjust stock market participation and the risky share of their financial portfolio over the life cycle?
- **Setting.** The paper studies Norwegian households from **1995–2009** using administrative tax-registry data that report complete household portfolios.
- **Core empirical challenge.** Existing evidence is hard to interpret because:
  - most portfolio data are cross-sectional, so age, cohort, and calendar-time effects are hard to separate;
  - the risky share is observed only for stock market participants, so selection into participation matters;
  - survey data on household portfolios are noisy and may contain age-correlated reporting error.

- **Main empirical contribution.** The paper documents a **double adjustment** over the life cycle:
  - before retirement, households who participate in the stock market gradually reduce the risky share;
  - after retirement, households increasingly exit the stock market altogether.
- **Main model contribution.** The paper extends a standard life-cycle portfolio model with:
  - uninsurable labor income risk;
  - borrowing constraints;
  - a fixed *per-period* participation cost for holding risky assets;
  - a small probability of a large stock market loss.
- **Main quantitative result.** The model fits the joint pattern best with:
  - relatively high risk aversion;
  - a small annual participation cost;
  - a yearly probability of a large stock market loss of about **1%**, close to the historical frequency of large Norwegian stock market crashes.
- **Economic takeaway.** Human capital explains why risky shares decline before retirement, but it does not explain why people leave the stock market after retirement. To generate both margins jointly, one needs a participation cost and a perceived probability of rare large losses.

## 2 Data and measurement (key objects)

### 2.1 Norwegian administrative portfolio data

- The core data come from the **Norwegian Tax Registry**. Because Norway levies a wealth tax, households report detailed financial and real asset holdings to the tax authority every year.
- Financial intermediaries send asset information directly to both households and the tax authority. This greatly reduces underreporting and measurement error.
- The authors draw a **20% random sample** of households in the 1995 population and follow them annually through **2009**.
- Main sample:
  - **164,015** households in 1995;
  - **1,804,115** household-year observations;
  - **106,369** households in the balanced panel.
- A household is defined around a married or cohabiting couple, and the household's age and cohort are measured using the husband's age and birth year.

## 2.2 Assets and portfolio definitions

- The paper observes detailed financial holdings, including:
  - bank deposits;
  - treasuries and bonds;
  - directly held stocks;
  - mutual funds;
  - money market funds.
- The authors collapse the financial portfolio into two broad asset classes:
  - **Risky assets:** directly held stocks plus mutual funds with a stock component;
  - **Safe assets:** bank deposits, money market funds, and bonds.
- **Financial wealth** is the sum of risky and safe financial assets.
- A household is defined as a stock-market participant if it holds at least **USD 160** in risky assets, measured in 1995 dollars.
- The **conditional risky share** is:

$$\alpha_{ia} = \frac{\text{risky financial assets}_{ia}}{\text{total financial wealth}_{ia}},$$

but this object is meaningful only for households that participate in risky asset markets.

## 2.3 Key descriptive facts from Table I

- In 1995, the average household head is about **51 years old**.
- Mean household financial wealth is about **USD 38,270**; median financial wealth is only about **USD 11,884**, showing substantial skewness.
- Mean net worth is about **USD 120,354**; a large fraction of wealth is real estate.
- Stock-market participation is limited:
  - **33%** participate in risky asset markets;
  - **23%** hold stocks directly;
  - **22%** hold mutual funds.
- Among risky-asset participants, the average risky share is about **32%**.
- Attrition is modest and largely mechanical: the average annual attrition rate is **3%**, and most exit occurs because one spouse dies.

## 2.4 Life-cycle objects measured in the paper

- The paper studies three central life-cycle profiles:

- participation in risky asset markets;
- conditional risky share among participants;
- financial wealth.
- It also studies entry and exit from risky asset markets using two definitions:
  - **Entry/exit**: moving into or out of risky asset ownership between age  $a$  and  $a + 1$ ;
  - **First-time entry/permanent exit**: entry by households that have never previously held risky assets, and exit by households that never reenter later.
- These entry/exit definitions help separate short-run portfolio switching from true life-cycle transitions into and out of risky asset markets.

## 2.5 Unobservables and timing

- Important unobservables include:
  - persistent and transitory labor income shocks;
  - unobserved participation costs or financial sophistication;
  - unobserved determinants of whether a household participates in the stock market;
  - cohort-specific financial experiences.
- The empirical timing is annual:
  - households begin a year with financial wealth and labor income;
  - they decide whether to participate in risky asset markets and how much to hold in risky assets;
  - portfolio returns and labor income shocks affect next-period wealth.
- In the structural model, participation is a forward-looking decision because paying a participation cost today changes exposure to the equity premium and disaster risk.

## 3 Models and empirical specifications (all equations + interpretation)

### 3.1 Empirical selection model for participation and risky share

The paper estimates a Heckman selection model because the risky share is observed only for participants.

The risky-share equation is:

$$s_{iact} = \beta_a A_a + \beta_c C_c + \beta_t D_t + \beta_0 Trend + \theta Z_{iact} + \theta_2 \lambda_{iact} + \varepsilon_{iact}.$$

The participation equation is:

$$\Pr(P_{iact} = 1 \mid x) = \Pr(P_{iact}^* > 0 \mid x)$$

$$= \Pr(\delta_a A_a + \delta_c C_c + \delta_t D_t + \delta_0 Trend + \vartheta Z_{iact} + \vartheta_2 L_{iact} + \eta_{iact} > 0).$$

Here:

- $s_{iact}$  is the conditional risky share of household  $i$  at age  $a$ , cohort  $c$ , and time  $t$ ;
- $P_{iact}$  is an indicator for participation in risky asset markets;
- $A_a$ ,  $C_c$ , and  $D_t$  are age, cohort, and calendar-year effects;
- $Z_{iact}$  includes controls such as family demographics and homeownership;
- $L_{iact}$  is lagged total lifetime wealth, measured as financial wealth plus estimated human wealth;
- $\lambda_{iact}$  is the inverse Mills ratio from the participation equation.

**Interpretation.** The participation decision and the risky-share decision are linked. Wealth strongly affects whether it is worthwhile to pay fixed participation costs, but conditional on participation, the risky share should depend mainly on age-related human-capital composition rather than the level of lifetime wealth. This gives the exclusion restriction used for the Heckman model.

### 3.2 Age, time, and cohort identification

A central difficulty is:

$$\text{calendar year} = \text{age} + \text{birth year}.$$

So age, time, and cohort effects cannot all be freely identified without additional restrictions.

The paper uses two strategies.

- **Deaton–Paxson restriction.** After detrending, time effects are restricted to sum to zero:

$$\sum_t \beta_t = 0, \quad \sum_t \delta_t = 0.$$

- **Cohort-proxy strategy.** Cohort effects are proxied by stock market returns experienced during formative years, especially ages **18–25**. This follows the idea that early macro-financial experiences shape beliefs and risk attitudes.

**Interpretation.** The paper does not rely on one arbitrary decomposition. It checks whether the core life-cycle profiles survive under two different ways of breaking the age-period-cohort collinearity.

### 3.3 Financial wealth profile

To estimate the life-cycle path of financial wealth, the paper uses:

$$z_{iact} = \gamma_a A_a + \gamma_c C_c + \gamma_t D_t + \psi Z_{iact} + \nu_{iact},$$

where:

- $z_{iact}$  is household financial wealth;

- $G_c$  measures macroeconomic conditions faced by cohort  $c$  when entering the labor market;
- wealth is adjusted for the correlation between mortality and wealth.

**Interpretation.** This equation is used to construct the empirical wealth profile that the structural model later tries to match. The mortality adjustment matters because older surviving households are selected on wealth.

### 3.4 Household objective in the structural model

Households work from age  $T_b$  to retirement age  $T_r$ , and then retire. They face survival risk. The objective is:

$$\mathbb{E} \sum_{a=T_b}^T \delta^a \left( \prod_{j=0}^{a-1} p_j \right) U(c_{ia}),$$

where:

- $\delta$  is the discount factor;
- $p_j$  is the probability of surviving to the next age;
- $U(c)$  is CRRA utility;
- $\gamma$  is the coefficient of relative risk aversion.

**Interpretation.** Survival risk lowers the effective discount factor at old ages. Households choose consumption, saving, participation, and risky-asset holdings while accounting for the probability of dying.

### 3.5 Financial markets and disaster risk

The risk-free return is constant:

$$r_f.$$

The ordinary risky return is:

$$r_t = r_f + r_p + \nu_t, \quad \nu_t \sim N(0, \sigma_r^2).$$

The paper then adds rare disaster risk:

$$r_t = \begin{cases} r_f + r_p + \nu_t, & \text{with probability } 1 - p_{tail}, \\ r_{tail}, & \text{with probability } p_{tail}. \end{cases}$$

Households also face a per-period participation cost  $q$  if they hold risky assets.

They cannot borrow against future labor income:

$$s_{ia} \geq 0, \quad b_{ia} \geq 0.$$

**Interpretation.** The disaster state lowers the attractiveness of stocks, especially for older households whose consumption depends more heavily on accumulated assets than on future labor income.

### 3.6 Budget constraints and labor income

For working households:

$$c_{ia} + \mathbf{1}_{i,a+1}(s_{i,a+1} + q) + b_{i,a+1} = w_{ia} + (1 + r_a)\mathbf{1}_{ia}s_{ia} + (1 + r_f)b_{ia}.$$

For retired households:

$$c_{ia} + \mathbf{1}_{i,a+1}(s_{i,a+1} + q) + b_{i,a+1} = \phi_{ret}w_{i,T_r} + (1 + r_a)\mathbf{1}_{ia}s_{ia} + (1 + r_f)b_{ia}.$$

During working age, log labor income follows:

$$\log w_{ia} = f(a) + v_{ia} + \eta_{ia},$$

with persistent component:

$$v_{ia} = v_{i,a-1} + u_{ia}.$$

During retirement:

$$\log w_{ia} = \log(\phi_{ret}) + f(T_r) + v_{i,T_r}.$$

**Interpretation.** Human capital is risky before retirement but largely deterministic afterward. As households age, the remaining stock of labor income falls, so they rebalance financial wealth toward safety.

### 3.7 Bellman equations and participation decision

For a participating household, the value is:

$$V_a^{in}(x, v) = \max_{c, s', b'} U(c) + \beta_{a+1} \mathbb{E}_{v', x'} V_{a+1}(x', v'),$$

where next-period financial wealth is:

$$x' = \begin{cases} (1 + r_f)b' + (1 + r)s', & \text{with probability } 1 - p_{tail}, \\ (1 + r_f)b' + (1 + r_{tail})s', & \text{otherwise.} \end{cases}$$

For a nonparticipating household:

$$V_a^{out}(x, v) = \max_{c, b'} U(c) + \beta_{a+1} \mathbb{E}_{v'} V_{a+1}(x', v'),$$

with:

$$x' = (1 + r_f)b'.$$



The household chooses whether to participate:

$$V_a(x, v) = \max\{V_a^{in}(x, v), V_a^{out}(x, v)\}.$$

The participation rule is:

$$\mathbf{1}'(x, v) = \begin{cases} 1, & \text{if } V_a^{in}(x, v) > V_a^{out}(x, v), \\ 0, & \text{otherwise.} \end{cases}$$

**Interpretation.** Participation is valuable because it gives access to the equity premium, but it is costly because it requires paying  $q$  and exposes the household to rare losses. This generates an age-dependent wealth threshold for stock-market participation.

### 3.8 Bequest motive

In one version of the model, households derive warm-glow utility from bequests:

$$V_a(x, v) = \frac{c^{1-\gamma}}{1-\gamma} + \delta \mathbb{E}_{v', x'} \left[ p_{a+1} V_{a+1}(x', v') + (1 - p_{a+1}) \frac{b}{1-\gamma} \left( \frac{x'}{b} \right)^{1-\gamma} \right].$$

**Interpretation.** The bequest motive helps the model match the empirical fact that households do not decumulate financial wealth very quickly after retirement.

### 3.9 Structural estimation by simulated moments

The structural parameters are estimated by simulated method of moments:

$$\hat{\kappa} = \arg \min_{\kappa} (\Gamma(\kappa) - \mu)' I (\Gamma(\kappa) - \mu),$$

where:

- $\kappa$  contains preference and participation parameters such as  $\beta$ ,  $\gamma$ ,  $q$ , and  $p_{tail}$ ;
- $\Gamma(\kappa)$  are model-simulated life-cycle moments;
- $\mu$  are the empirical life-cycle moments estimated in the data;
- $I$  is the weighting matrix.

**Interpretation.** The model is judged by whether it can jointly match participation, conditional risky shares, and in the bequest version, financial wealth over the life cycle.

## 4 Identification and instruments (what makes the estimates credible)

### 4.1 What is hard to identify

- The key empirical objects are life-cycle profiles, but age, calendar time, and cohort move together mechanically.
- Risky shares are selected outcomes because they are observed only for stock-market participants.
- Portfolio patterns can reflect many mechanisms:
  - human-capital rebalancing;
  - fixed participation costs;
  - beliefs about rare disasters;
  - cohort-specific experiences;
  - wealth accumulation and mortality selection.
- The structural parameters are therefore not identified by one statistic. They are identified by the joint shape of several life-cycle profiles.

### 4.2 What makes the data credible

- The Norwegian tax-registry data contain nearly complete financial portfolios rather than survey self-reports.
- Financial institutions report asset information directly to the tax authority, so reporting error is much smaller than in household surveys.
- The panel is long enough to follow households through meaningful portions of the life cycle.
- The authors can observe both extensive-margin behavior, participation and exit, and intensive-margin behavior, the risky share among participants.

### 4.3 What identifies the empirical life-cycle profiles

- **Selection correction.** Lagged lifetime wealth shifts participation because the fixed cost of participation matters more for poor households. Conditional on age and other controls, it is excluded from the risky-share equation.
- **Age-period-cohort strategy.** The paper uses both a Deaton–Paxson restriction and a cohort-proxy strategy based on youth stock-market returns.
- **Entry and exit measures.** The distinction between first-time entry, temporary exit, and permanent exit helps reveal that young households often enter and exit temporarily, while old households increasingly exit permanently.
- **Mortality correction.** Wealth profiles are corrected because poorer households have different mortality patterns than richer households.

#### 4.4 What identifies the structural parameters

- The **participation cost** is mainly identified by limited participation and the wealth threshold for entering risky asset markets.
- The **risk aversion parameter** is mainly identified by the level of the conditional risky share.
- The **tail-event probability** is identified by the timing and speed of stock-market exit after retirement and by the need to lower risky shares without relying on very large participation costs.
- The **discount factor** helps control asset accumulation. Lower discount factors limit liquid wealth accumulation and therefore discourage participation.
- The **bequest motive** helps match the weak financial-wealth decumulation observed among retirees.

#### 4.5 Empirical credibility and limitations

- The structural tail-risk parameter is plausible because it is close to the historical frequency of Norwegian stock market crashes.
- The disaster loss is calibrated using actual household risky-portfolio losses in the 2008 crash: mean loss about **48.5%**.
- Norway is a special environment: unemployment and health risks are heavily insured relative to many countries. This helps isolate stock-market tail risk, but it may limit external validity.
- The model matches the broad double-adjustment pattern, but it still has imperfections:
  - it generates too little young-age participation;
  - it generates too fast exit later in life;
  - even with bequests, it predicts more retirement decumulation than observed.

#### 4.6 Relation to the literature

- The paper connects directly to Merton-style life-cycle portfolio theory, in which human capital generates a motive to reduce stock exposure with age.
- It contributes to household finance by using administrative data to document life-cycle patterns more cleanly than survey-based studies.
- It relates to participation-cost models by showing that a per-period cost is useful but not sufficient on its own.
- It also connects to rare-disaster and experience-based-belief ideas by showing that a small probability of a large stock loss is crucial for matching retirement-age exit.

## 5 Tables and figures

### 5.1 Table I: Descriptive statistics

Read:

- Table I establishes the quality and scale of the data: 164,015 households in the full sample and 106,369 in the balanced panel.
- Financial wealth is highly skewed: mean financial wealth is about USD 38,270, while the median is about USD 11,884.
- Participation in risky assets is limited at baseline: about 33% of households participate.
- Conditional on participation, the average risky share is about 32%.
- The balanced panel looks similar to the full sample in asset holdings and participation, suggesting that attrition is not driving the main results.

Table I  
Descriptive Statistics, 1995

This table displays summary statistics for the main sample of married households in the first year of observation, 1995. In addition, the table provides summary statistics for the sample of households that remain in the panel throughout the sample period, which ends in 2009. Where applicable, values are reported in 1995 USD. Education is missing for less than 1% of the sample.

|                                  | Full Sample<br>N = 164,015 |         |        | Balanced Panel Sample<br>N = 106,369 |         |        |
|----------------------------------|----------------------------|---------|--------|--------------------------------------|---------|--------|
|                                  | Mean                       | Std Dev | Median | Mean                                 | Std Dev | Median |
| <i>Demographics:</i>             |                            |         |        |                                      |         |        |
| Age husband                      | 50.88                      | 14.14   | 49     | 47.67                                | 11.64   | 47     |
| Age wife                         | 48.12                      | 14.01   | 47     | 45.00                                | 11.40   | 45     |
| Share less high school education | 0.22                       | 0.40    | 0      | 0.18                                 | 0.39    | 0      |
| Share high school education      | 0.53                       | 0.50    | 1      | 0.55                                 | 0.49    | 1      |
| Share college education          | 0.24                       | 0.44    | 0      | 0.27                                 | 0.43    | 0      |
| Household size                   | 3.24                       | 1.19    | 3      | 3.44                                 | 1.17    | 3      |
| <i>Asset holdings in USD:</i>    |                            |         |        |                                      |         |        |
| Financial wealth                 | 38,270                     | 106,975 | 11,884 | 38,169                               | 111,865 | 11,348 |
| Stocks                           | 12,797                     | 91,438  | 0      | 14,386                               | 97,230  | 0      |
| Mutual funds                     | 1,173                      | 3,895   | 0      | 1,245                                | 3,989   | 0      |
| Safe assets                      | 24,297                     | 37,678  | 9,734  | 22,536                               | 35,575  | 9,139  |
| Net worth                        | 120,354                    | 143,051 | 97,543 | 116,213                              | 142,199 | 93,318 |
| <i>Participant share:</i>        |                            |         |        |                                      |         |        |
| Risky assets                     | 0.33                       | 0.47    | 0      | 0.35                                 | 0.48    | 0      |
| Stocks                           | 0.23                       | 0.42    | 0      | 0.25                                 | 0.43    | 0      |
| Mutual funds                     | 0.22                       | 0.41    | 0      | 0.23                                 | 0.42    | 0      |
| <i>Mean share participants:</i>  |                            |         |        |                                      |         |        |
|                                  | N = 54,519                 |         |        | N = 37,770                           |         |        |
| Risky assets                     | 0.32                       | 0.30    | 0.20   | 0.33                                 | 0.31    | 0.22   |
| Stocks                           | 0.23                       | 0.31    | 0.05   | 0.24                                 | 0.32    | 0.06   |
| Mutual funds                     | 0.09                       | 0.15    | 0.03   | 0.09                                 | 0.15    | 0.04   |
| <i>Attrition:</i>                |                            |         |        |                                      |         |        |
|                                  | N = 58,863                 |         |        |                                      |         |        |
| Share death                      | 0.62                       |         |        |                                      |         |        |
| Share migration                  | 0.13                       |         |        |                                      |         |        |
| Share divorce/separation         | 0.25                       |         |        |                                      |         |        |
| Mean yearly attrition rate       | 0.03                       |         |        |                                      |         |        |
| Age at exit                      | 62.63                      | 16.83   | 64     |                                      |         |        |



Figure 1. Participation rates in risky asset markets. This figure plots the mean participation rates in risky asset markets by age for selected cohorts over the period 1995 to 2009.

### 5.2 Figure 1: Participation rates by cohort

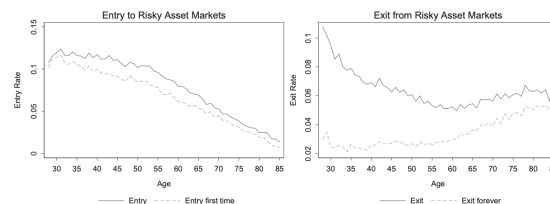
Read:

- The figure plots raw participation rates by age for selected birth cohorts.
- Young cohorts enter risky asset markets quickly as they age.
- Older cohorts have lower participation at the same age than younger cohorts, showing the importance of cohort effects.
- All cohorts are affected by common time shocks, such as market booms and recessions.

- The figure motivates the need to separate age, time, and cohort effects formally.

### 5.3 Table II and Figure 2: Entry and exit

| Table II<br>Measures of Stock Market Entry and Exit |                                                                                                                                   |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Measure 1                                           |                                                                                                                                   |
| Entry:                                              | The fraction of households that do not hold stocks at age $a$ that enter risky asset markets at $a+1$ .                           |
| Exit:                                               | The fraction of households that are stockholders at age $a$ that exit the market at age $a+1$ .                                   |
| Measure 2                                           |                                                                                                                                   |
| Entry:                                              | The fraction of households that have never held any stocks up until age $a$ that enter risky asset markets at $a+1$ .             |
| Exit:                                               | The fraction of households that are stockholders at age $a$ that exit the market at age $a+1$ and never reenter the stock market. |

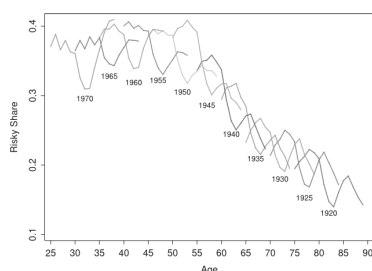


**Figure 2. Entry and exit rates to/from risky asset markets.** These figures plot entry and exit rates to/from risky asset markets. The left panel depicts entry and exit frequencies, allowing for reentry/exits, whereas the right panel depicts frequencies of first-time entry and once-and-for-all exits.

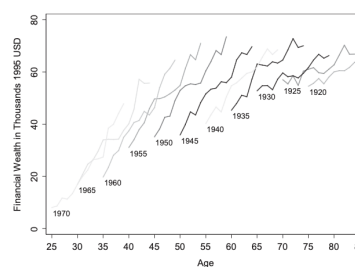
Read:

- Table II defines two versions of entry and exit: ordinary entry/exit and first-time entry/permanent exit.
- Figure 2 shows first-time entry is highest early in life, peaking around 13%.
- Permanent exit is low early in life but rises sharply after retirement.
- Temporary entry and exit are common among younger households, which suggests that participation is not a once-and-for-all decision.
- This supports the model choice of a *per-period* participation cost rather than only a one-time entry cost.

### 5.4 Figure 3 and Figure 4: Risky share and financial wealth by cohort



**Figure 3. Risky share of financial wealth by cohort.** This figure plots the average risky shares of households' financial portfolios conditional on participation, for selected cohorts by age.



**Figure 4. Financial wealth by cohort.** This figure plots the average financial wealth for selected cohorts by age. Values are in 1995 USD, and the wealth profiles are corrected for mortality following Attanasio and Hoynes (2000), as outlined in Appendix B.

Read:

- Figure 3 shows that conditional risky shares are high at young ages and then decline as households approach retirement.
- Figure 4 shows that financial wealth rises strongly through working life.
- There is only mild financial-wealth decumulation after retirement.

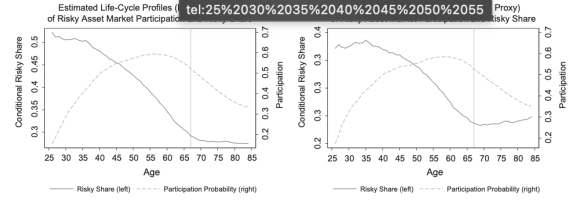
- Together, these figures show that portfolio adjustment is not simply mechanical decumulation of all assets. Households change both whether they hold stocks and how much of their portfolio they allocate to stocks.

## 5.5 Table III and Figure 5: Heckman estimates of participation and risky share

**Table III**  
**Heckman Selection Model**

This table displays the two estimated Heckman (1979) selection models (discussed in Section B) for asset market participation and the conditional risky share. Lagged total wealth is the sum of financial and human wealth (in 100,000 of 1995 USD), and  $\lambda$  is the inverse Mills ratio/nonselection hazard. Coefficients in the selection equation are calculated marginal effects of the underlying probit regression. For presentational purposes, calendar year fixed effects and family size coefficients are reported in Section VI of the Internet Appendix; age coefficients and marginal effects are displayed in Figure 5. Standard errors are in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

|                        | Deaton-Paxson        |                      | Cohort Proxy         |                   |
|------------------------|----------------------|----------------------|----------------------|-------------------|
|                        | (1)<br>Participation | (2)<br>Share         | (3)<br>Participation | (4)<br>Share      |
| Trend                  | 0.012***<br>(0.001)  | -0.003***<br>(0.001) |                      |                   |
| Youth stock return     |                      |                      | 0.361***<br>(0.017)  | -0.070<br>(0.080) |
| Lag total wealth       | 4.107***<br>(0.148)  |                      | 4.186***<br>(0.030)  |                   |
| $\lambda_{\text{act}}$ |                      | -0.186***<br>(0.001) | -0.186***<br>(0.001) |                   |
| Observations           | 1,804,115            | 886,189              | 1,804,115            | 886,189           |
| Joint sign. tests      |                      |                      |                      |                   |
| Year $\chi^2$ (12)     | 1,575.79***          | 882.70***            | 1,488.21***          | 698.54***         |
| Cohort $\chi^2$ (59)   | 7,644.51***          | 19.17***             |                      |                   |



**Figure 5. Estimation: risky asset market participation and risky share.** The left panel in the figure plots the life-cycle patterns for both risky asset market participation and the conditional risky share of financial wealth, which are estimated using the Heckman selection equation applying the Deaton and Paxson (1994) methodology reported in columns (1) and (2) of Table III. The right panel applies the cohort-proxy methodology reported in columns (3) and (4) of Table III. For the selection/participation equation, we plot the marginal values of the estimated underlying probit equation, and for the risky share, the age coefficients of the outcome equation in the Heckman model.

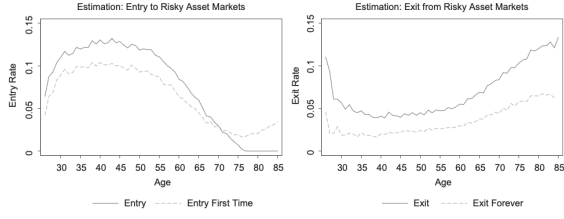
### Read:

- Table III shows that lifetime wealth strongly predicts participation, consistent with fixed participation costs.
- The inverse Mills ratio in the risky-share equation is significant and negative, which means selection into participation matters.
- Youth stock returns significantly predict participation under the cohort-proxy specification.
- Figure 5 is the core empirical figure:
  - participation is hump-shaped, rising until middle/late working age and falling after retirement;
  - the conditional risky share starts high, declines before retirement, and stabilizes after retirement.
- The most important message is the timing: the risky share adjusts before retirement, while stock-market exit begins mainly after retirement.

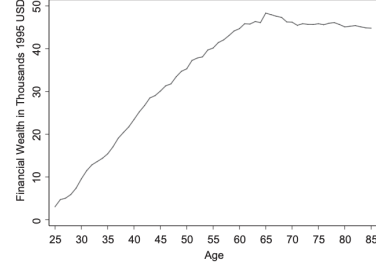
## 5.6 Figure 6 and Figure 7: Entry, exit, and wealth profiles

### Read:

- Figure 6 confirms that entry and exit are life-cycle phenomena after accounting for age, cohort, and time effects.
- Entry is hump-shaped and peaks around middle age.
- Exit is U-shaped, with especially high exit at older ages.



**Figure 6. Life-cycle patterns of entry to and exit from risky asset markets.** These figures plot the estimated life-cycle patterns of entry to and exit from risky asset markets, defined by the two sets of measures in Table II imposing the Deaton-Paxson (1994) restriction of zero-sum time effects (see Section B).



**Figure 7. Life-cycle pattern of financial wealth.** This figure plots the estimated life-cycle pattern of financial wealth when correcting for differences in mortality, following Attanasio and Hoynes (2000), and proxying for cohort effects with deviations from trend GDP-per-capita at early ages (Kapteyn, Alessie, and Lusardi (2005), Jappelli (1999)). Values are in 1995 USD.

**Table IV  
Parameterization**

| Parameter                              |                | Value   | Source                               |
|----------------------------------------|----------------|---------|--------------------------------------|
| Retirement age                         | $T^r$          | 67      | Norwegian Law                        |
| Risk-free return                       | $r_f$          | 0.0143  | Klovland (2004),                     |
| Risk premium                           | $r_p$          | 0.0314  | Dimson, Marsh, and Staunton (2008) & |
| Stock return std deviation             | $\sigma_r$     | 0.238   | Ødegaard (2007)                      |
| Tail event return                      | $r_{tail}$     | -0.485  | Table C.II                           |
| Variance of transitory shocks          | $\sigma_t^2$   | 0.023   | Table A.I                            |
| Variance of persistent shocks          | $\sigma_p^2$   | 0.012   | Table A.I                            |
| Replacement ratio                      | $\phi_{ret}$   | 0.842   | Table A.II                           |
| Shape of Pareto distribution for $x_0$ | $\mu_{x_0}$    | 0.452   | Wealth at age 25                     |
| Scale of Pareto distribution for $x_0$ | $\sigma_{x_0}$ | 5,711.7 | Wealth at age 25                     |

- Figure 7 shows the estimated financial-wealth profile: wealth rises sharply earlier in life and decumulates only mildly after retirement.
- This weak retirement decumulation motivates the bequest version of the model.

## 5.7 Table IV: Parameterization

Read:

- Table IV lists externally calibrated parameters.
- Retirement age is set to **67**, consistent with Norwegian law.
- The annual risk-free return is **1.43%**; the equity premium is **3.14%**; stock-return volatility is **23.8%**.
- The tail-event return is set to **-48.5%**, based on household losses during the 2008 crash.
- The pension replacement ratio is estimated at about **0.842**.
- The table makes clear that the model is tightly connected to Norwegian data rather than only generic calibration targets.

## 5.8 Figure 8 and Figure 9: Policy functions and participation thresholds

Read:

- Figure 8 shows optimal risky shares as a function of wealth at different ages.
- The risky share declines with wealth and age because bond-like human capital becomes less important as people age or accumulate financial assets.

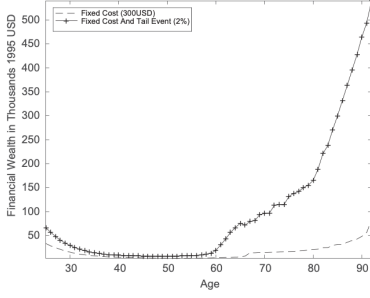


Figure 9. Wealth threshold of participation as a function of age.

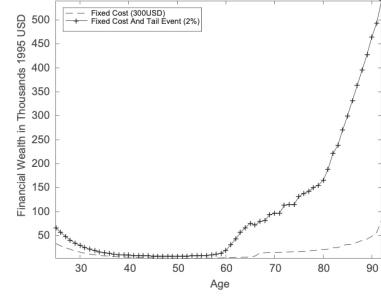


Figure 9. Wealth threshold of participation as a function of age.

- Adding the participation cost creates a wealth threshold for stock-market participation.
- Figure 9 shows that this participation threshold is mildly U-shaped over age.
- Adding disaster risk raises the participation threshold, especially at older ages.
- The key mechanism: retirees rely more on financial wealth for consumption, so tail losses are more dangerous for them.

## 5.9 Figure 10: Model simulations

Read:

- The baseline Cocco–Gomes–Maenhout model predicts 100% stock-market participation at all ages.
- Adding a per-period participation cost generates limited participation and some exit, but the exit occurs too late and is too weak.
- Adding tail risk produces more retirement-age exit while also lowering the conditional risky share.
- The model captures the qualitative ordering of adjustments: risky-share rebalancing starts before stock-market exit.
- However, the model still does not fully match the flat risky share early in life.

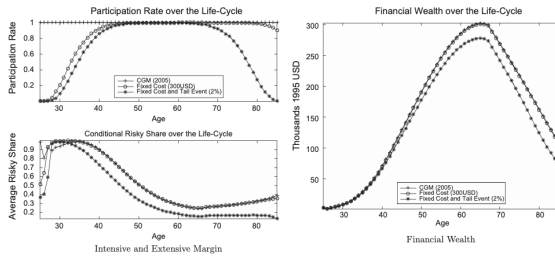


Figure 10. Model simulations.

Table V  
Structural Estimation of Parameters

In Estimations 1 and 2, we target the average participation rate and the average conditional risky share of households between ages 26 and 85. In Estimation 3, we augment the targeted moments with the average financial wealth of households between ages 26 and 85. More details on the estimation can be found in the Internet Appendix. The targeted data moments are those estimated using the Deaton-Paxson methodology in Section II.

|                                                     | Estimation 1         | Estimation 2         | Estimation 3         |
|-----------------------------------------------------|----------------------|----------------------|----------------------|
| Discount factor ( $\beta$ )                         | 0.827<br>(0.03e-06)  | 0.753<br>(0.08e-06)  | 0.770<br>(0.01e-06)  |
| Risk aversion ( $\gamma$ )                          | 14.444<br>(3.05e-06) | 11.762<br>(1.50e-06) | 11.011<br>(0.69e-06) |
| Participation cost ( $q$ )                          | 0.344<br>(0.31e-06)  | 0.109<br>(0.06e-06)  | 0.065<br>(0.05e-06)  |
| Probability of tail event ( $p_{tail}$ )            | —                    | 0.011<br>(0.01e-06)  | 0.010<br>(0.01e-06)  |
| Value of objective function ( $L(\hat{\epsilon})$ ) | 1,136.32             | 865.52               | 1,406.88             |
| Nr. of targeted moments ( $M$ )                     | 118                  | 118                  | 177                  |
| Wealth to income ratio, age 65                      | 3.76                 | 1.67                 | 1.98                 |
| Wealth to income ratio, age 65 (data)               | 1.62                 | 1.62                 | 1.62                 |



## 5.10 Table V: Structural estimation

Read:

- Table V is the main structural table.
- Without tail risk, Estimation 1 requires very high risk aversion,  $\gamma = 14.444$ , and a participation cost of about USD 344 per year.
- Allowing tail risk in Estimation 2 lowers risk aversion to  $\gamma = 11.762$  and the participation cost to about USD 109 per year.
- The estimated tail probability is **1.1%** in Estimation 2.
- With a bequest motive and wealth-profile targeting in Estimation 3, risk aversion falls to  $\gamma = 11.011$ , the participation cost falls to about USD 65, and the tail probability is **1.0%**.
- The tail probability estimates are close to the historical crash frequency in Norway, which strengthens the model's interpretation.

## 5.11 Figure 11 and Figure 12: Structural fit

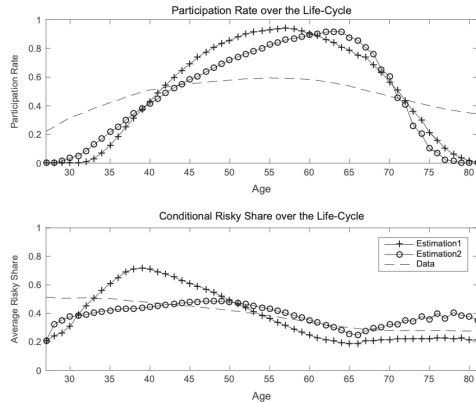


Figure 11. Estimations 1 and 2.

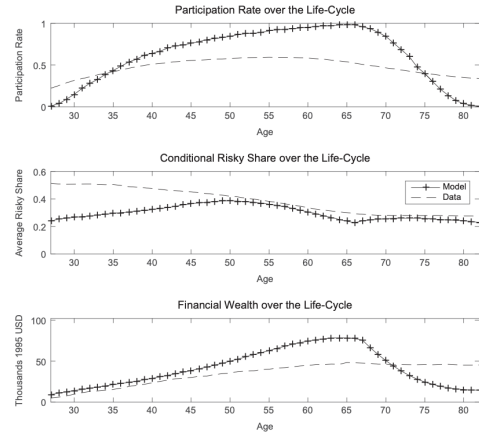


Figure 12. Estimation 3.

Read:

- Figure 11 compares the model with and without tail risk to the empirical participation and risky-share profiles.
- Adding tail risk improves the timing of stock-market exit and lowers the conditional risky share to a more realistic level.
- Figure 12 shows the model with bequest motive targeting participation, risky share, and financial wealth.
- The model tracks financial wealth fairly well until retirement but still predicts too much decumulation at very old ages.

- The figures show that the model is successful in matching the broad pattern, but it remains too stylized to match every age-specific detail.

## 5.12 Appendix tables: Labor income and crash calibration

**Table A.I**  
**Income Variance Decomposition and Correlation with Stock Returns**  
This table reports estimates of the variance of permanent and transitory labor income shocks. The estimation is based on the error terms from estimating the labor income process in Figure A.1. The procedure is based on the method in Carroll and Samwick (1997), which is also used in Cocco, Gomes, and Maenhout (2005). *t*-values are in parentheses.

|              | < High School    | High School      | College          | All              |
|--------------|------------------|------------------|------------------|------------------|
| Transitory   | 0.026<br>(9.97)  | 0.015<br>(6.76)  | 0.029<br>(11.89) | 0.023<br>(16.5)  |
| Permanent    | 0.013<br>(17.44) | 0.011<br>(19.12) | 0.011<br>(17.73) | 0.012<br>(31.11) |
| Stock market | 0.017<br>(0.48)  | 0.045<br>(0.34)  | 0.005<br>(0.75)  | 0.008<br>(0.50)  |

**Table A.II**  
**Labor Income Process: Age Polynomials**  
The table shows the coefficients of the fifth-order polynomial that approximates labor income as a function of age for various population groups.

|                         | < High School | High School | College | All    |
|-------------------------|---------------|-------------|---------|--------|
| Age                     | 0.042         | 0.038       | 0.112   | 0.057  |
| Age <sup>2</sup> /10    | -0.023        | -0.018      | -0.069  | -0.031 |
| Age <sup>3</sup> /100   | 0.006         | 0.004       | 0.021   | 0.009  |
| Age <sup>4</sup> /1000  | -0.101        | -0.091      | -0.337  | -0.161 |
| Age <sup>5</sup> /10000 | 0.075         | 0.074       | 0.208   | 0.117  |
| Constant                | 3.576         | 3.657       | 3.415   | 3.566  |
| Observations            | 61            | 61          | 61      | 61     |
| <i>r</i> <sup>2</sup>   | 0.995         | 0.996       | 0.988   | 0.995  |

Read:

- Table A.I estimates the variance of labor income shocks. Permanent and transitory labor-income risk are modest in Norway, partly because administrative data reduce measurement error and social insurance cushions shocks.
- The correlation between labor income and stock returns is small and statistically insignificant, so labor income does not provide a strong hedge against stock market risk.
- Appendix C documents the frequency and size of Norwegian stock-market crashes.
- During the 2008 crash, the mean household risky-portfolio loss is **48.5%**, which pins down  $r_{tail}$ .

**Table C.I**  
**Large Stock Market Drops Experienced by Households**

The table displays the distribution of the number of large stock market drops (in the upper panel drops of more than 25% in a year, and in the lower panel drops of more than 30%) experienced by the households in the sample for various age groups: above the age of 18, between ages 18 and 36, between ages 18 and 42, and between ages 18 and 55.

| Number of Drops | Experience of Medium/Large Drops $\geq 25\%$ |                   |                   |                   |
|-----------------|----------------------------------------------|-------------------|-------------------|-------------------|
|                 | Total Life-Cycle                             | Age Bracket 18–36 | Age Bracket 18–42 | Age Bracket 18–55 |
| 0               | 1.52                                         | 39.97             | 24.89             | 5.45              |
| 1               | 2.28                                         | 18.85             | 8.77              | 6.22              |
| 2               | 8.90                                         | 41.01             | 63.25             | 39.21             |
| 3               | 22.04                                        | 0.17              | 3.09              | 24.03             |
| 4               | 17.49                                        |                   |                   | 19.46             |
| 5               | 47.77                                        |                   |                   | 5.63              |
| Mean            | 3.9                                          | 1.0               | 1.4               | 2.66              |
| Median          | 4                                            | 1                 | 2                 | 3                 |

| Number of Drops | Experience of Large Drops $\geq 30\%$ |                   |                   |                   |
|-----------------|---------------------------------------|-------------------|-------------------|-------------------|
|                 | Total Life-Cycle                      | Age Bracket 18–36 | Age Bracket 18–42 | Age Bracket 18–55 |
| 0               | 4.37                                  | 53.36             | 35.40             | 8.36              |
| 1               | 17.99                                 | 46.47             | 61.52             | 54.42             |
| 2               | 29.88                                 | 0.17              | 3.09              | 31.59             |
| 3               | 47.77                                 |                   |                   | 5.63              |
| Mean            | 2.2                                   | 0.5               | 0.7               | 1.3               |
| Median          | 2                                     | 0                 | 1                 | 1                 |

**Table C.II**  
**Net Returns on Risky Assets in 2008**

The table gives summary statistics of the cross-sectional distribution of net returns on households' risky asset portfolios in 2008

| Percentile | Net Return |
|------------|------------|
| 1st        | -70%       |
| 5th        | -58.6%     |
| 10th       | -56.4%     |
| 25th       | -55.3%     |
| 50th       | -47.8%     |
| 75th       | -47.0%     |
| 90th       | -46.7%     |
| 95th       | -41.0%     |
| 99th       | -29.8%     |
| Mean       | -48.5%     |
| Range      | -40.2%     |
| Std Dev    | 6.0%       |

## 6 Short summary

- **Method.** The paper combines Norwegian administrative portfolio data with a selection-corrected empirical life-cycle analysis and a structural life-cycle portfolio model.

- **Empirical finding.** Households adjust along two margins: they reduce risky shares before retirement and exit the stock market after retirement.
- **Identification.** The empirical design deals with selection into stockholding and separates age, time, and cohort effects using both Deaton–Paxson restrictions and cohort proxies based on early-life stock returns.
- **Structural result.** A per-period participation cost generates limited participation, but a small probability of a large stock-market loss is needed to match the timing of exit after retirement and the level of risky shares.
- **Main takeaway.** Human capital explains pre-retirement rebalancing, while the interaction of participation costs and tail risk explains post-retirement exit.

## 7 Conclusion

### 7.1 Core conclusion

- The paper provides strong evidence that households do rebalance away from stocks as they approach retirement, contrary to earlier claims that risky shares are roughly flat over the life cycle.
- It also documents a second, distinct margin: older households increasingly leave the stock market after retirement.
- Standard life-cycle portfolio models can explain the first margin, but not the second.
- Extending the model with per-period participation costs and rare disaster risk can explain the broad double-adjustment pattern.

### 7.2 Economic intuition

- Young households have a large stock of future labor income, which behaves partly like a safe asset. This makes stockholding attractive and supports high risky shares.
- As retirement approaches, human capital shrinks. Households compensate by lowering the risky share of financial wealth.
- After retirement, households rely more heavily on accumulated financial wealth. A large stock-market crash would then have especially severe consumption consequences.
- Because older households also choose lower risky shares, the equity premium is less valuable relative to the participation cost. This makes exit more attractive.

### 7.3 Takeaway

The paper shows that life-cycle portfolio choice is not just about the risky share conditional on participation. Participation itself is a central life-cycle margin. Administrative data reveal a clear timing pattern: households rebalance before retirement and exit after retirement. A model with

human capital, per-period participation costs, and rare large losses provides a coherent explanation of this timing.